MILITARY SPECIFICATION

SHAKING MACHINE, LABORATORY, Blood Collecting Tube

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a mechanical shaking machine, suitable for use in thoroughly mixing, by rotation, blood samples contained in blood collecting tubes, in medical laboratories.

1.2 Classification. Shaking machine covered by this specification shall be of the following types (see 6.1):

Type I. - Angular wheel.
Type II. - Vertical wheel.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Fiberboard, Shipping.

MILITARY

MIL-P-116 - Preservation-packaging, Methods of.
MIL-M-7298 - Manual, Technical, Commercial Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Fed. Std. No. 66 - Steel: Chemical Composition and Hardenability.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73-11 Dimensions of Plugs and Receptacles-
General Purpose, 125 Volts, 15 Amperes,
2 Pole, 3 Wire, Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Material. Material shall be suitable for the purpose intended and shall be as hereinafter specified. Material not definitely specified shall be of the best quality used for the purpose in commercial practice.

3.2 Construction. Shaking machine shall consist of a housing which contains a motor, a wheel and a shaft. Shaft shall connect the motor to the wheel.

3.2.1 Type I.

3.2.1.1 Motor housing. Motor housing sides and top shall be fabricated of carbon steel sheet. Steel sheet for sides and top shall be U.S. Standard Revised Guage No. 20, 0.0359 inch thick. Base shall be fabricated from aluminum which is at least 3/16 inch thick. An "off-on" toggle switch shall be located on the right side of the housing. The housing shall have suitably located openings which provide ventilation during operation. The front of the housing shall be slanted inward to provide for angular mounting of the wheel. Base shall be secured to the sides with screws and shall provide a sturdy and stable foundation for the shaker. Rubber feet shall be secured to each of the four corners of the base to afford minimum vibration and prevent creepage during operation.

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3.2.1.2 Wheel. Wheel shall be fabricated of carbon steel sheet. Steel sheet shall be U.S. Standard Revised Gauge No. 20, 0.0359 inch thick. The wheel shall be mounted on front of the housing at $30^{\circ} \pm 3^{\circ}$ angle from the vertical. Twelve tube-retaining clamps (see 3.2.5) shall be placed equidistantly around the circumference of the face 1 ± 1/16 inch from the edge of the wheel. Clamps shall be fabricated of spring steel and shall be secured to the wheel by nuts, bolts and lock washers.

3.2.2 Type II.

3.2.2.1 Motor housing. Motor housing shall be fabricated of corrosion-resisting steel complying with Fed. Std. 66, AISI 302 or 304. Steel for sides and top shall be U.S. Standard Revised Gauge No. 26, 0.0179 inch thick. Steel for base shall be U.S. Standard Revised Gauge No. 22, 0.0299 inch thick. An "off-on" toggle switch shall be located on the right hand side. The housing shall have suitably located openings which provide ventilation during operation. Three sides shall be secured to the base by spot welding. One side and the top shall be removable to provide for accessibility to the motor. Wheel shall be mounted on the front in vertical position. Distance from the edge of housing to the edge of the base shall be as follows: front - 2 inches minimum, sides-equidistant and back of housing shall not extend beyond the edge of the base.

3.2.2.2 Wheel. Wheel shall be fabricated of methyl methacrylate. Wheel shall have twelve slots which extend from near the edge toward the center. Slots shall be at a $20^{\circ} \pm 3^{\circ}$ angle from the vertical with the deepest portion of each slot toward the center of the wheel. Blood collecting tubes shall fit (see 3.2.5) and shall be held in the slots by a spring wire which encircles the periphery of the slots and by an aluminum disk which shall be secured to the center of the wheel. Disk shall be equipped with a friction slip clutch which makes it possible to insert or remove a tube without stopping the wheel. Rubber feet shall be secured to each of the four corners of the base.

3.2.3 Motor. The rotating wheel shall be turned by a fractional horsepower motor which requires no lubrication. Motor shall drive the wheel at a speed between 15 and 24 revolutions per minute.

3.2.4 Power cord and plug. A three-conductor line cord with a three-prong male plug conforming to ANSI C73.11 shall be provided. Length of cord shall be 6 feet (minimum). At least 5-3/4 feet of cord shall extend outside the motor housing.

3.2.5 Performance. Construction of shaking machine shall be such that it operates in a smooth and steady manner without undue vibration. The 12 and 16 millimeter tubes shall not be displaced more than 1/8 inch from the original placement when tested as specified in 4.5.1.

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3.2.6 Dimensions. Dimensions shall be as specified in table I.

TABLE I. Dimensions in inches.

Type I	Dimension
Motor housing	
Height	9
Width	8-1/2
Depth	6-1/8
Wheel	
Diameter	11-3/8
Type II	
Motor housing	
Height	6-3/8
Width, upper portion	5-1/16 (Length of front and back)
Depth, upper portion	4-9/16 (Length of sides)
Width, base	6-5/8
Depth, base	7-5/8
Wheel	
Diameter	4-3/4
Thickness	1
Aluminum disk diameter	3-1/2

3.2.7 Tolerances. Material gauges specified shall be considered the minimum acceptable gauges (see 3.2.1.1 & 3.2.2.1). Commercial tolerances shall apply to gauge thickness. Tolerances in dimensions shown in table I shall be ± 1/8 inch.

3.3 Finish. Type I shaking machine shall have a hammertone finish on the exposed surfaces and a smooth enamel finish on the inside surfaces. The inside and outside surfaces of the base shall have an enamel finish. All surfaces of the motor housing of the type II shaking machine shall have a No. 4 commercial finish. Screws, bolts, nuts and washers shall be fabricated from corrosion resisting metal or shall be coated to resist corrosion.

3.4 Identification marking. A nameplate shall be permanently attached to the shaking machine. Nameplate shall be legibly marked with the electrical characteristics, wattage, voltage, frequency and type of current and the name or registered trademark of the manufacturer.

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3.5 Service data. The contractor shall furnish, with each unit, two copies of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.6 Workmanship. Workmanship shall be first class throughout. Shaking machine shall be free of defects listed in table IV and any other defects which detract from the appearance or may impair the serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. For examination.

	Inspection level	AQL (defects per 100)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III. Acceptance number shall be zero.

TABLE III. For tests.

Component	Characteristic	Requirement	Test procedure	Inspection level
Shaking machine	Performance test	3.2.5	4.5.1	S-1

4.3 Examination. Shaking machine shall be examined to determine compliance with all requirements contained in this specification. Examination shall be in accordance with, but shall not be restricted to, the defects listed in table IV.

TABLE IV. Classification of defects.

Categories	Defects
Major	
101	Component part missing.
102	Service data missing.
103	Shaking machine not free of wobbling or rocking when placed on flat surface.
104	Shaking machine not free of burrs or sharp edges.
105	Shaking machine not free of rust spots or corroded areas.
106	Shaking machine not free of indentations or protrusions.
107	Finish not free of chipping, cracking or peeling (type I).
108	Component parts not secure.

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TABLE IV. Classification of defects. (Continued)

Categories	Defects
Major	
109	Rubber feet supports, and rubber washers not free of cracks, tears or tackiness.
110	Methyl methacrylate wheel not free of cracks, chips breaks or deformation (type II).
111	Steel spring not free of deformation (type II).
112	Wheel not rotating smoothly or not free of wobble (type II).
Minor	
201	Finish not free of blisters, or rough areas (type I).
202	Unit not free of pits and scratches.
203	Nameplate not attached securely or permanently.
204	Marking on nameplate not correct, legible or complete.
205	Methyl methacrylate wheel not smoothly finished (type II).
206	Methyl methacrylate wheel not free of scratches, pits or protrusions.
207	Wire cord not flexible.
208	Spring-retaining clamps not equidistantly positioned on surface of wheel (type I).

4.3.1 Dimensional examination. The shaking machine shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Performance test. The wheel of the unit shall be loaded with 12 millimeter (mm.) diameter tubes filled with water to a 3/4 capacity point and the shaking machine allowed to operate a minimum of 20 minutes. At the end of the cycle, any displacement of the tubes from their original position shall be noted. This procedure shall be repeated using 16 mm. diameter tubes.

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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. Each type I or type II shaking machine, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Shaking machine shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.2.2 Drying. Shaking machine shall be dried in accordance with MIL-P-116.

5.1.2.3 Preservative application. Preservatives shall not be used with the exception of preservatives applied to internal moving parts as deemed necessary by the contractor.

5.1.2.4 Unit packaging. Each shaking machine shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.3 Level C. Shaking machine shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Each shaking machine of the type specified shall be packed in an exterior container designed for a type 1 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Each shaking machine of the type specified shall be packed in an exterior container designed for a type 1 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged shaking machine shall be packed in shipping containers that will afford adequate protection against damage during shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking. Each unit and exterior container shall be marked in accordance with MIL-Std-129.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Type required (see 1.2).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item as listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6640-00-982-4243	SHAKING MACHINE, LABORATORY, Blood Collecting Tube, 115 volt, 60 Hz, AC

6.3 Limitations. This specification does not cover all types of the commodity indicated by the title of this specification or those which are commercially available, but is intended to cover the types normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing acticity:
DLA-DM

Project No. 6640-0978

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-S-37453 SHAKING MACHINE, LABORATORY, Blood Collecting Tube

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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